

Structural Bioinformatics

Oral Exam Preparation

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To do list (in order of importance)

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1. SB08 Comparative Modelling
2. SB10 Ab Initio Methods
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4. SB14 Molecular Dynamics
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6. SB04 Protein Folding
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8. SB05 Structural Data
9. SB12-SB13 Disorder
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1 SB01 – Introduction to Structural Bioinformatics

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1. Definition of bioinformatics
2. Structural bioinformatics
3. Central dogma of molecular biology
4. Protein structure and function
5. Structure prediction problem
6. Sequence–structure gap
7. Protein folding problem
8. AlphaFold and AI-based prediction
9. Course objectives
10. Computational tools and applications

Possible questions:

1. What is structural bioinformatics?
2. How does structural bioinformatics differ from classical bioinformatics?
3. What is the sequence–structure gap?
4. Why is protein structure important for understanding function?
5. What is the protein folding problem?
6. Why was protein structure prediction historically difficult?
7. How did AlphaFold change the field?
8. What disciplines contribute to bioinformatics?
9. What are the main applications of structural bioinformatics?
10. What computational skills are important in this field?

2 SB02 – Chemical Bonds

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1. Matter and chemical transformations
2. Atomic theory
3. Atomic structure
4. Electrons and orbitals
5. Chemical bonding principles
6. Covalent bonds
7. Ionic interactions
8. Electronegativity
9. Molecular geometry
10. Non-covalent interactions in biomolecules

Possible questions:

1. What is the difference between physical and chemical transformations?
2. Describe the structure of an atom.
3. What are orbitals and quantum numbers?
4. Explain the Aufbau principle.
5. What is electronegativity?
6. Compare covalent and ionic bonds.
7. What determines bond polarity?
8. Why are non-covalent interactions important in proteins?
9. What role do hydrogen bonds play in biology?
10. Why is carbon particularly suited for life?

3 SB03 – Amino Acids / Acid-Base Theory

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1. Arrhenius theory
2. Brønsted–Lowry theory
3. Acids and bases in aqueous solution
4. Strong vs weak acids
5. Acid dissociation constant
6. pKa
7. pH and pOH
8. Water self-ionization
9. Amino acid ionization
10. Buffer systems

Possible questions:

1. Compare Arrhenius and Brønsted–Lowry theories.
2. What is a conjugate acid-base pair?
3. Define K_a and pKa.
4. What is the relationship between pKa and acid strength?
5. What is the difference between a strong acid and a weak acid?
6. What is the significance of pH?
7. Why is water amphoteric?
8. What is a buffer?
9. Explain the Henderson–Hasselbalch equation.
10. How do amino acids behave at different pH values?
11. What is the isoelectric point of a protein?

4 SB04 – Protein Folding

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1. Protein folding problem
2. Native and denatured states
3. Levinthal's paradox
4. Secondary structure
5. Folding pathways
6. Energy landscape
7. Thermodynamics of folding
8. Folding kinetics
9. Chaperones
10. Misfolding and aggregation

Possible questions:

1. What is the native state of a protein?
2. What is the protein folding problem?
3. What is the typical folding timescale in biological proteins?
4. What is Levinthal's paradox?
5. What are the main secondary structures?
6. What are the backbone dihedral angles?
7. What is the Ramachandran plot?
8. How can we describe the folding process in terms variations of thermodynamic variables?
9. What is the hydrophobic effect?
10. What stabilizes the native state?
11. Difference between thermodynamics and kinetics of folding?
12. Why is protein folding not a random search?
13. Explain the energy funnel model.
14. What is the principle of minimal frustration?
15. What is residual frustration and why is it important?
16. What are molecular chaperones?
17. How is misfolding related to disease?

What is the protein folding problem? Most proteins in biological systems are functional only when they are folded into a specific 3D structure, called the native state. The process by which a protein acquires its native structure is called folding. The protein folding problem is the challenge of understanding both **which** is the 3d structure that the protein will fold into (the native structure) starting from the sequence, and **how** it will get there, meaning which sequence of conformations will it sample, and if there is a specific, dominant pathway that it will follow, or many.

So the protein folding problem has mainly two parts: one is the **structure prediction problem**, which is the question that methods like AlphFold are trying to answer, and the other is the **folding mechanism problem**, which is the question of how the protein gets to its native state, and regards the kinetics of folding.

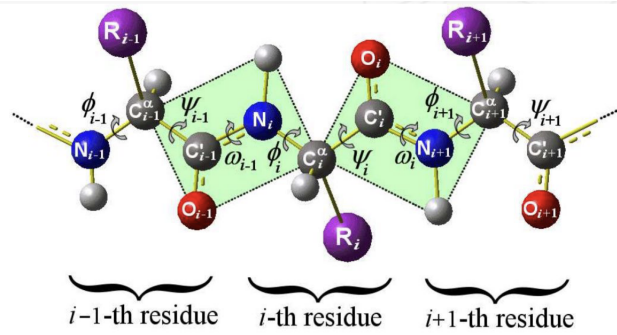
The structure prediction problem is framed like this: given the aminoacid sequence, find the **injective function** that maps the sequence to the 3D structure. Methods like alphafold try to learn this function from the data, and they are very successful at it. However, they do not give us any insight into the folding mechanism problem, which is still an open question. Also, these methods are intrinsically limited by the data they are trained on: for instance, they only predict with a water environment, cannot say anything about what happens when the protein is in a different solvent, and they saw only the 20 L-aminoacids in the training data, so they cannot predict what happens when there are post-translational modifications, or non-natural aminoacids.

What is the typical folding timescale in biological proteins? Most biological proteins fold spontaneously on timescales ranging from microseconds to seconds, although the exact folding rate depends on protein size, topology, and stability. Small single-domain proteins can fold in microseconds or milliseconds, whereas larger proteins may require seconds or the assistance of molecular chaperones.

What is Levinthal's paradox? Levinthal's paradox is the observation that proteins fold into their native structure much faster than would be possible if they explored all possible conformations randomly. Cyrus Levinthal estimated that even a small protein could adopt an astronomically large number of conformations, making a random search of conformational space effectively impossible within the age of the universe. However, experimental observations show that most proteins fold in times ranging from microseconds to seconds. This apparent contradiction suggests that protein folding is not a random search process. Instead, proteins fold through biased pathways guided by an energy landscape that directs the molecule toward the native state. Additionally, biological proteins are not random sequences, but have been selected through evolution to fold in timescales compatible with life.

What are the main secondary structures? Secondary structure refers to local, regularly recurring arrangements of the protein backbone **stabilized primarily by hydrogen bonds** between backbone amide and carbonyl groups. The two most common secondary structures are the α -helix and the β -sheet. An α -helix is a right-handed helical structure in which each carbonyl oxygen forms a hydrogen bond with the amide hydrogen four residues ahead in the sequence ($i \rightarrow i + 4$). A β -sheet consists of extended β -strands connected by hydrogen bonds and can be either parallel or antiparallel. The antiparallel arrangement is the most stable because it allows atoms that participate in the hydrogen bond to get closer.

What are the backbone dihedral angles?



The bond $N - C_\alpha - C_\beta$ is rigid and planar. Therefore, the backbone conformation of a protein can be described by two dihedral angles: ϕ (phi) and ψ (psi). The ϕ angle is the rotation around the $N-C_\alpha$ bond and is defined by the three atoms $(C_\beta^{-1}, N, C_\alpha)$, while the ψ angle is the rotation around the $C_\alpha - C_\beta$ bond and is defined by the three atoms $(C_\alpha, C_\beta, N^{+1})$.

What is the Ramachandran plot? The Ramachandran plot is a graphical representation of the backbone dihedral angles ϕ and ψ . Each point on the plot corresponds to a particular combination of these two angles, allowing visualization of the conformational space accessible to the protein backbone. Because of steric repulsion between atoms, many combinations of ϕ and ψ are energetically unfavorable and therefore forbidden. As a result, residues cluster in a few allowed regions corresponding to common secondary structures. For example, right-handed α -helices and β -sheets occupy distinct areas of the Ramachandran plot. The Ramachandran plot is widely used to understand protein structure and to assess the quality of experimentally determined or predicted structures. A high-quality protein model should have most residues located in energetically allowed regions, while residues in disallowed regions may indicate structural errors or unusual conformations.

How can we describe the folding process in terms of variations of thermodynamic variables? Protein folding can be described using the Gibbs free energy change

$$\Delta G_{fold} = \Delta H_{fold} - T\Delta S_{fold}.$$

Folding is thermodynamically favorable when $\Delta G_{fold} < 0$, meaning that the folded state has a lower free energy than the unfolded state. The enthalpy term ΔH_{fold} is generally negative because folding creates stabilizing interactions such as hydrogen bonds, van der Waals contacts, and salt bridges.

The entropy change has two opposing contributions:

$$\Delta S_{fold} = \Delta S_{conf} + \Delta S_{solvent}$$

The protein loses conformational freedom upon folding, leading to a decrease in conformational entropy ($\Delta S_{conf} < 0$), which opposes folding. However, hydrophobic residues become buried in the protein core, releasing ordered water molecules into the solvent and increasing solvent entropy ($\Delta S_{solvent} > 0$), which favors folding.

Overall, the sign of ΔG_{fold} depends on the balance between these enthalpic and entropic contributions. Usually it is negative, meaning that the folded state is thermodynamically favored. But not by a lot: it is said that proteins are "marginally stable": the free energy difference between the folded and unfolded states is typically only a few kcal/mol (5 – 15), which is small compared to the total energy of the protein (for a comparison, a single hydrogen bond accounts for 1 – 5 kcal/mol). This marginal stability allows proteins to be flexible and dynamic, enabling them to perform their biological functions effectively.

What is the hydrophobic effect? The hydrophobic effect is the tendency of nonpolar molecules or side chains to cluster together in aqueous environments. In proteins, hydrophobic residues such as Val, Leu, Ile, Phe, and Met are typically buried in the protein core, while polar and charged residues remain exposed to the solvent. The driving force behind the hydrophobic effect is not a direct

attraction between hydrophobic groups. Instead, it arises from the behavior of water molecules. When hydrophobic residues are exposed, surrounding water molecules become highly ordered to maintain their hydrogen-bonding network. This ordering decreases the entropy of the solvent. Upon folding, hydrophobic residues are buried, reducing the exposed hydrophobic surface and releasing these ordered water molecules into the bulk solvent, thereby increasing solvent entropy. The hydrophobic effect is one of the major driving forces of protein folding.

Why is protein folding not a random search? Because it cannot be. The Levital's paradox assumed that every residue can sample all possible conformations independently. But this is not true, there are constraints on the conformational space that the protein can sample, such as steric hindrance and the formation of local secondary structures. These constraints significantly reduce the number of conformations that need to be explored. Also, the folding process is guided by an energy landscape that directs the protein towards the native state. As some of the native interaction form, the space of possibilities gets restricted and other native interactions become more likely to form, and this goes on till the protein reaches the minimum of the energy landscape, which is the native state.

Explain the energy funnel model view and how it differs from the ant trail view The funnel model essentially says that at the start of the folding process, the protein can sample a wide variety of conformations, but as it folds and forms native interactions, the conformational space narrows down. This is a funnel-shaped energy landscape, where the bottom represents the native state. The folding process is stochastic: usually there is no "single" pathway dominating over the others. Different molecules of the same protein may follow different routes and visit different intermediates, yet still reach the same native structure.

What is the principle of minimal frustration? Frustration means that different interactions "disagree" about what the structure should be. Minimal frustration means that most interactions "agree" that the native state is the best solution. This concept is tightly connected to the energy funnel model: the funnel exists precisely because evolution has minimized frustration in naturally occurring protein sequences.

The principle of minimal frustration states that natural proteins have evolved sequences in which most interactions favor the formation of the native structure rather than competing with it. As a result, the energy landscape is biased toward the native state and contains relatively few energetic conflicts, or *frustrations*, that could trap the protein in incorrect conformations. It is overall a quite smooth landscape. This greatly reduces the number of kinetic traps and allows proteins to fold rapidly and reproducibly.

What is residual frustration and why is it important? Although natural proteins obey the principle of minimal frustration, frustration is not completely eliminated. *Residual frustration* refers to the small number of interactions within a folded protein that are not fully optimized for structural stability and that locally favor alternative conformations or competing interactions. Residual frustration is important because proteins are not evolved solely for stability. Many functional processes, such as ligand binding, allosteric regulation, conformational changes, and enzymatic catalysis, require a certain degree of flexibility and structural adaptability. Locally frustrated regions often coincide with active sites, binding interfaces, or regions involved in molecular recognition. Therefore, while minimal frustration enables efficient folding by creating a funnel-shaped energy landscape, residual frustration provides the flexibility necessary for biological function. Proteins can be viewed as a compromise between the requirements of stability and the requirements of activity.

How is misfolding related to disease? Protein misfolding occurs when a protein fails to reach its native structure or adopts an alternative conformation that is unstable or non-functional. Misfolded proteins may lose their biological activity, become degraded, or aggregate with other misfolded molecules. Because protein function depends strongly on structure, misfolding can have severe cellular consequences. Many diseases are associated with the accumulation of misfolded proteins and the

formation of insoluble aggregates or amyloid fibrils. Examples include Alzheimer's disease, which involves aggregation of amyloid- β peptides; Parkinson's disease, associated with α -synuclein aggregates; and prion diseases, in which a misfolded protein induces the misfolding of other copies of the same protein. In these cases, protein aggregation can disrupt cellular processes and lead to cell death. Cells possess quality-control mechanisms, including molecular chaperones and degradation pathways, that help proteins fold correctly and remove misfolded species. Disease can arise when these systems are overwhelmed or when mutations increase the tendency of a protein to misfold and aggregate.

5 SB05 – Structural Data

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1. Experimental determination of protein structures
2. X-ray crystallography
3. Nuclear Magnetic Resonance (NMR)
4. Cryo-Electron Microscopy (Cryo-EM)
5. Protein Data Bank (PDB)
6. PDB file formats and annotations
7. Structure quality and resolution
8. Biological assemblies and complexes
9. Structural databases and resources
10. Applications of structural data

Possible questions:

1. What is the Protein Data Bank (PDB)?
2. Which information is contained in a PDB structure file?
3. What experimental techniques are used to determine protein structures?
4. What is the difference between X-ray crystallography, NMR spectroscopy and cryo-EM?
5. What is meant by resolution in structural biology?
6. What is a biological assembly and how does it differ from the asymmetric unit?
7. What are "missing residues" in a PDB structure and why do they occur?
8. Why is structural annotation important?
9. What are PDB-derived databases and why are they useful?
10. How can structural data contribute to drug discovery and protein engineering?
11. What are the main limitations and biases of currently available structural datasets?
12. How has the growth of the PDB enabled the development of comparative modelling and AlphaFold?
13. Why is structure determination still important despite highly accurate structure prediction methods?
14. How are experimental structures used to derive statistical potentials and train machine learning models?

6 SB06 – Structural Alignments

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1. Structural comparison of proteins
2. Structural superposition
3. Protein structure representations
4. Root Mean Square Deviation (RMSD)
5. Kabsch algorithm
6. Global and local structural alignment
7. Alignment scoring and significance
8. Structural similarity measures
9. Conservation of structure versus sequence
10. Applications of structural alignments
 1. What is RMSD?
 2. How is RMSD calculated?
 3. What is the difference between structural alignment and structural superposition?
 4. What is the Kabsch algorithm?
 5. Why are structural alignments more difficult than sequence alignments?
 6. What information is lost when using only C α atoms?
 7. What are equivalent residues in a structural alignment?
 8. Why can structure be more conserved than sequence?
 9. What are the limitations of RMSD as a similarity measure?
 10. When is structural comparison useful in bioinformatics?

7 SB07 – Structural Evolution

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1. Protein evolution
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3. Structural conservation
4. Homology and analogy
5. Divergent evolution
6. Convergent evolution
7. Protein folds and fold space
8. Structural classification systems
9. Remote homology detection
10. Evolutionary constraints on protein structure
 1. Why is protein structure generally more conserved than sequence?
 2. What is the relation between sequence similarity and structural similarity in biological proteins?
 3. What level of sequence identity is usually associated with structural similarity?
 4. What are homologous proteins?
 5. What is divergent evolution?
 6. What is convergent evolution?
 7. Can proteins with low sequence identity have similar structures?
 8. What is a protein fold?
 9. Explain the Rossmann fold as an example of structural conservation.
10. How does evolution constrain protein structure?
11. Why is structural information useful for detecting remote homology?

8 SB08 – Comparative Modelling

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1. Protein structure prediction
2. Principles of homology modelling
3. Template identification
4. Sequence alignment
5. Model building
6. Loop modelling
7. Side-chain modelling
8. Model refinement
9. Model validation and quality assessment
10. Limitations of comparative modelling
 1. What is comparative (homology) modelling?
 2. Why do similar sequences generally adopt similar structures?
 3. What is the role of the template in homology modelling?
 4. Describe the complete homology-modelling pipeline.
 5. Why is sequence alignment the most critical step?
 6. Why are loops difficult to model accurately?
 7. How are side chains modelled?
 8. How is model quality evaluated?
 9. What sequence identity is generally considered reliable for modelling?
 10. What are the main limitations of comparative modelling?

9 SB09 – Statistical Potentials

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1. Protein scoring functions
2. Native versus non-native structures
3. Knowledge-based potentials
4. Distance-dependent statistical potentials
5. RAPDF
6. Bayesian formulation
7. Likelihood and prior probabilities
8. Structure evaluation
9. Fold recognition
10. Advantages and limitations of statistical potentials
 1. What is a statistical potential?
 2. How are statistical potentials derived from structural databases?
 3. What is RAPDF?
 4. Why are native structures distinguishable from non-native structures?
 5. What is the role of Bayes' theorem in statistical potentials?
 6. What assumptions underlie RAPDF?
 7. How are residue-residue distances used in scoring functions?
 8. What is the difference between physical and statistical potentials?
 9. What applications do statistical potentials have in structural bioinformatics?
 10. What are the main limitations of statistical potentials?

10 SB10 – Ab Initio Methods

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1. CASP experiments
2. Categories of structure prediction methods
3. Template-free prediction
4. Fragment assembly methods
5. Conformational search
6. Energy functions
7. Sampling strategies
8. Prediction assessment
9. AlphaFold and deep learning approaches
10. Current challenges and future directions

Possible Exam Questions:

1. What is CASP and why is it important? ✓
2. What does the term 'ab initio prediction' mean? ✓
3. How are predicted structures evaluated in CASP? ✓
4. What is ROSETTA and how does it work? ✓
5. How is the ROSETTA scoring potential modelled? What differs between the "generation" and "evaluation" steps? ✓
6. When the potential energy has several solutions with comparable low energy, how does ROSETTA assign the native one and why? ✓
7. Why is the global minimum broader than any other minimum - both in the *real* energy landscape and in the ROSETTA scoring function landscape? ✓
8. How did AlphaFold change the field of structure prediction?
9. What is the main difference between AlphaFold and traditional contact-prediction methods based on covariance analysis?
10. Why does AlphaFold predict inter-residue distance distributions rather than simple residue contacts?
11. What are the main inputs used by AlphaFold2, and what information is contained in a Multiple Sequence Alignment (MSA)?
12. How does AlphaFold2 exploit evolutionary information contained in homologous sequences?
13. What is the Evoformer, and what role does it play in AlphaFold2?
14. What is the attention mechanism, and why is it particularly suitable for protein structure prediction?
15. What is the purpose of the MSA representation and the pair representation in AlphaFold2?
16. What is triangular self-attention, and how does it help enforce geometric consistency in protein structures?
17. What is Invariant Point Attention (IPA), and why is it necessary for generating 3D coordinates?
18. How does the structure module convert Evoformer representations into atomic coordinates?
19. What is the recycling procedure in AlphaFold2, and why does it improve prediction accuracy?
20. What training data were used to train AlphaFold2? What is self-distillation?
21. What are the main outputs produced by AlphaFold2 besides the predicted structure?
22. What is pLDDT, and how should it be interpreted when evaluating a prediction?
23. What is the Predicted Aligned Error (PAE), and what information does it provide that pLDDT does not?
24. What physical or geometric knowledge is explicitly incorporated into AlphaFold2?
25. Why is AlphaFold2 considered an end-to-end system?
26. What are the main strengths of AlphaFold2?
27. What are the main limitations of AlphaFold2? Why does it struggle with conformational vari-

- ability, protein complexes, mutations, and ligand binding?
28. How does AlphaFold differ from Rosetta in its approach to protein structure prediction?
 29. Why are ab initio methods computationally expensive?
 30. What role do energy functions play in structure prediction?
 31. How is conformational space explored in ab initio prediction?
 32. What challenges remain in protein structure prediction?

What does the term 'ab initio prediction' mean? The term *ab initio* (Latin for “from the beginning”) refers to methods that predict the three-dimensional structure of a protein starting only from its amino acid sequence, without relying on experimentally determined homologous structures as templates, and more in general without relying on the knowledge of other protein structures in general.

What is CASP and why is it important? CASP (Critical Assessment of Structure Prediction) is a community-wide experiment that evaluates the performance of protein structure prediction algorithms. The organizers distribute only the amino acid sequences of proteins whose experimental structures are not yet public. Participants submit predictions, and after a few months the results are discussed at a conference. CASP has been held every two years since 1994 and today comprises several categories in addition to structure prediction, including disorder prediction and model quality assessment. A particularly important subcategory of structure prediction is **free modelling or "ab initio" modelling**, where no reliable template (homologous structure) is available.

How are predicted structures evaluated in CASP? The main metric used in CASP to evaluate the accuracy of predicted structures is the Global Distance Test Total Score (GDT-TS). GDT-TS measures the percentage of residues in the predicted structure that are within a certain distance threshold from their corresponding residues in the experimentally determined structure. The score ranges from 0 to 100, with higher scores indicating better predictions.

What is Rosetta and how does it work? Rosetta is a template-free method that dominated the CASP competition for years, until it was surpassed by AlphaFold. It is based on the idea of **fragment assembly**. Works as follows:

1. Split the sequence in segments of 9 residues,
2. Use MSA to find fragments of known structures that have similar sequence profiles,
3. Assemble these fragments (coming from unrelated proteins) into a complete structure,
4. Define a potential energy function -or *scoring function*- that combines physical and statistical terms:

$$P(\text{structure}|\text{sequence}) = P(\text{structure}) \times P(\text{sequence}|\text{structure})$$

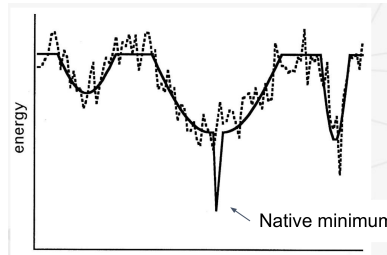
5. Perform simulated annealing on this potential surface to find the global minima,
6. Cluster together similar conformations of low energy,
7. Select the broadest cluster as the native ensemble.

Although Rosetta is often classified as an ab initio or de novo method, it is not a purely ab initio approach because it relies on fragment libraries and statistical information extracted from previously solved protein structures. Its main innovation is the combination of fragment assembly, Monte Carlo sampling, and knowledge-based scoring functions to efficiently explore conformational space.

How does Rosetta select the native conformation when it finds several structures with similar low energy? Rosetta typically generates thousands of candidate structures (called *decoys*) during the fragment assembly and Monte Carlo search process. Often, many of these structures have very similar energies, making it difficult to identify the native conformation based solely on the energy score. After generating the decoys, Rosetta groups structurally similar models into clusters and identifies the **most populated cluster**. The representative structure of the largest low-energy cluster is often selected as the final prediction. This strategy is based on the observation that random incorrect (although low in energy) conformations are usually scattered throughout conformational

space, whereas native-like structures tend to accumulate in the same region. In other words: **the native minimum is broader than any other minimum**. Thus, Rosetta does not simply choose the structure with the absolute lowest energy; instead, it combines **energy ranking** with **clustering analysis**, selecting structures that are both low in energy and frequently observed during the search. This improves robustness against inaccuracies in the *scoring function* and increases the probability of identifying the native fold.

Why is the global minimum broader than any other minimum - both in the *real* energy landscape and in the ROSETTA scoring function landscape?



In the *real energy landscape*, the native minimum is expected to be broader than other minima because the stability of the native state is dominated by long-range hydrophobic interactions. Hydrophobic residues tend to cluster in the protein core, and this favorable packing can be maintained even when the structure undergoes small conformational changes. As a result, many conformations that are similar to the native structure have comparable energies, creating a wide basin in the energy landscape. The native conformation corresponds to a sharp, deep drop in this surrounding basin of low energy.

Rosetta uses an approximate scoring function based on statistical potentials and simplified physical models. This scoring function captures broad features such as hydrophobic burial quite well, but is less accurate in describing more specific interactions such as hydrogen bonds, electrostatic interactions, and van der Waals contacts. Consequently, the sharp energy drop associated with the exact native structure is partially obscured by noise. However, it is still able to reconstruct the *broad region* surrounding the native state, where many low-energy conformations accumulate. Therefore, native-like structures tend to form large clusters in the *rosetta* conformational space, whereas incorrect folds are usually represented by isolated minima with fewer nearby low-energy conformations.

How is ROSETTA scoring potential modelled? What differs between the "generation" and "evaluation" steps? Rosetta uses a scoring function based on Bayes' theorem to estimate the probability that a candidate structure is the correct fold for a given sequence: $P(\text{structure}|\text{sequence}) \propto P(\text{structure})P(\text{sequence}|\text{structure})$. It applies a two-step process: a **generation step** that produces candidate structures using a simplified scoring function, and an **evaluation step** that assesses these candidates with a more detailed scoring function. During structure generation, Rosetta requires a scoring function that is extremely fast because millions of candidate conformations must be evaluated. Therefore it uses a simplified statistical potential:

$$P(\text{structure}|\text{sequence}) \approx e^{-R_g^2} \prod_{i < j} \frac{P(r_{ij}|aa_i, aa_j)}{P(r_{ij})}$$

which contains two main terms:

1. Radius of gyration term $P(\text{structure}) \sim e^{-R_g^2}$: which favors compact conformations and discriminates folded proteins from random coils. This depends on structure only, i.e. it is a prior distribution.
2. Pairwise distance potential: $\frac{P(r_{ij}|aa_i, aa_j)}{P(r_{ij})}$: which measures how likely it is to observe two amino acids at distance r_{ij} . This is a statistical term.

Importantly, the simplified model assumes independence between residue pairs and ignores environmental effects, i.e. whether the residue pair is solvent-exposed or buried. Once a smaller set of candidate structures has been generated, Rosetta can afford a more accurate scoring function which accounts for

the environment: the potential has a different shape if the residue pair is buried in the protein core or exposed to solvent.

Transcript

Rosetta Rosetta was one of the dominant methods for protein structure prediction before AlphaFold.

The main idea is that the set of protein structures observed in the PDB can be recombined to generate plausible new structures. Protein structures are built by assembling short fragments taken from known structures.

Fragment selection The target sequence is divided into short segments.

For each segment, Rosetta searches the PDB for similar sequence fragments. Instead of using only the raw sequence, it uses **sequence profiles**, obtained from multiple sequence alignments (MSAs).

A sequence profile describes the frequency of each amino acid at each position of the alignment.

For each position, Rosetta selects a set of candidate fragments (typically the top 25 fragments from the PDB).

Fragment assembly The next problem is to find the best combination of fragments.

The search space is enormous, so Rosetta uses stochastic sampling methods:

- discard conformations with steric clashes;
- use Monte Carlo-like sampling;
- allow some unfavorable moves to escape local minima.

The idea is to explore the energy landscape and avoid becoming trapped in local minima.

11 Statistical Potentials in Rosetta

Instead of calculating physical energies exactly using molecular mechanics or molecular dynamics, Rosetta often uses **statistical potentials**, which are much faster.

The goal is to estimate:

$$P(\text{structure} \mid \text{sequence})$$

Using Bayes' theorem:

$$P(\text{structure} \mid \text{sequence}) = \frac{P(\text{sequence} \mid \text{structure})P(\text{structure})}{P(\text{sequence})}$$

Since all candidate structures correspond to the same sequence, $P(\text{sequence})$ can be ignored:

$$P(\text{structure} \mid \text{sequence}) \propto P(\text{sequence} \mid \text{structure})P(\text{structure})$$

Radius of gyration The probability of a structure can be related to its compactness.

The **radius of gyration** measures how spread out the atoms are around the center of mass. Folded proteins are generally much more compact than random coils.

Environment-dependent amino acid preferences Rosetta evaluates how likely it is to observe a given amino acid in a given structural environment.

Examples include:

- alanine in an α -helix;
- hydrophobic residues buried in the protein core;
- charged residues exposed to solvent.

Pairwise distance potentials Rosetta also evaluates the probability of observing two amino acids (or atoms) at a given distance.

This is the basis of pairwise statistical potentials such as DFIRE or RAPDF-like potentials.

A limitation is that these potentials often assume independence between residue pairs.

12 Environment-Dependent Pair Potentials

A more sophisticated potential considers not only the distance between two residues, but also their environments.

For example, the interaction between two residues may differ depending on whether they are:

- both buried;
- both solvent-exposed;
- one buried and one exposed.

Hydrophobic residues Hydrophobic residues are generally favorable at short distances, particularly when buried in the protein core.

At sufficiently large distances their contribution should approach zero because they no longer interact.

12.1 Oppositely charged residues

Opposite charges are generally attractive at short distances, especially when forming salt bridges.

Same-charge residues Same-charge residues are usually unfavorable at short distances because of electrostatic repulsion.

However, in specific environments (e.g. ion-binding sites or catalytic sites), same-charge residues can occur close together.

Adding environmental information allows statistical potentials to better reproduce realistic physical interactions.

13 Model Selection and Clustering

Rosetta generates a very large number of candidate structures.

Native-like conformations often form large clusters in conformational space.

Therefore, a useful strategy is to select structures that are:

- low in energy;
- members of large structural clusters.

This reflects the idea that the native state lies within a broad favorable basin of the energy landscape.

Foldit Foldit is a protein-folding game that allows users to manipulate protein structures manually.

The software evaluates moves using force fields and scoring functions.

The original idea was to combine human spatial intuition with computational energy evaluation.

AlphaFold

13.1 AlphaFold in CASP

AlphaFold revolutionized protein structure prediction.

AlphaFold 1 appeared at CASP13 (2018) and surpassed previous methods.

AlphaFold 2 produced a major breakthrough at CASP14.

Many targets achieved GDT-TS values around 90.

A GDT-TS score of approximately 90 indicates that most residues are predicted very close to their experimentally determined positions.

14 Covariation and Multiple Sequence Alignments

A key idea behind modern structure prediction is **coevolution**.

If two residues interact in a protein structure, a mutation in one residue may require a compensatory mutation in the other residue to preserve the interaction.

This creates correlated mutations in a multiple sequence alignment.

Therefore, sequence information alone can reveal which residues are likely to be close in three-dimensional space.

Early contact-prediction methods typically required roughly 1000 homologous sequences to obtain reliable contact maps.

AlphaFold 1 AlphaFold 1 uses:

- the target sequence;
- a multiple sequence alignment.

It predicts:

- residue-residue distance distributions;
- torsion-angle distributions.

Rather than predicting coordinates directly, it predicts geometric constraints and then builds structures satisfying those constraints.

AlphaFold 2 AlphaFold 2 is an **end-to-end neural network**.

Inputs:

- target sequence;
- multiple sequence alignment;
- optional structural templates.

Output:

- full three-dimensional coordinates.

The most important components are:

- the Evoformer;
- the Structure Module.

The Evoformer The Evoformer processes two main representations.

14.1 MSA Representation

Contains information extracted from the multiple sequence alignment.

It captures evolutionary correlations between columns.

Pair Representation Contains information describing relationships between residue pairs.

The Evoformer allows these two representations to exchange information.

Important operations include:

- attention over MSA rows and columns;
- transfer of covariation information into pair representations;
- triangular updates.

Triangular updates exploit geometric consistency between residue triplets and allow inference of additional residue contacts.

15 Structure Module

The Structure Module converts internal representations into atomic coordinates.

Residues are represented as rigid bodies that initially occupy the same position.

The network predicts:

- rotations;

- translations;
- side-chain torsion angles.

A major innovation is **Invariant Point Attention (IPA)**.

IPA allows the network to reason directly in three-dimensional space while remaining invariant to global rotations and translations.

After coordinate generation, a final relaxation step can be applied to remove clashes and improve stereochemistry.

16 Confidence Measures

AlphaFold provides confidence estimates in addition to coordinates.

16.1 pLDDT

The predicted Local Distance Difference Test (pLDDT) is a per-residue confidence score.

- High pLDDT: high confidence.
- Low pLDDT: uncertain or potentially disordered regions.

16.2 PAE

The Predicted Aligned Error (PAE) estimates the expected positional error between residue pairs.

It is especially useful for multidomain proteins.

A model may predict each domain accurately while remaining uncertain about their relative orientation.

17 Self-Distillation

AlphaFold 2 employed **self-distillation**.

The procedure was:

1. train an initial model on experimentally determined PDB structures;
2. predict structures for a large set of proteins from UniRef;
3. select high-confidence predictions;
4. add these predictions to the training set.

This substantially improved model performance.

18 Strengths of AlphaFold

- Predicts structures directly from sequence.
- Often works even without close templates.
- Exploits evolutionary information from MSAs.
- Produces highly accurate predictions.
- Provides useful confidence estimates.

19 Limitations of AlphaFold

- Limited handling of post-translational modifications.
- Not reliable for predicting mutation effects.
- Usually predicts a single dominant conformation.
- Struggles with conformational changes induced by ligands, pH, ions, etc.
- Limited treatment of protein complexes (partially addressed by AlphaFold-Multimer).
- Does not naturally model DNA, RNA, membranes, or most ligands.
- Antibody–antigen interactions remain challenging.
- Performance depends on available evolutionary information and training data.

AlphaFold is not a molecular dynamics simulation and does not reproduce the physical folding pathway. It predicts the most likely final structure.

20 Main Conceptual Difference Between Rosetta and AlphaFold

Rosetta and AlphaFold represent two different paradigms.

Rosetta: Fragment assembly, Monte Carlo sampling, and statistical potentials derived from known structures.

AlphaFold: End-to-end deep learning using sequence information, evolutionary covariation, attention mechanisms, and geometric reasoning.

The transition from Rosetta to AlphaFold represents the shift from manually designed scoring functions and sampling procedures to learned representations of protein sequence, evolution, and structure.

21 SB11 – Tandem Repeat Proteins

Table of Contents

1. Tandem repeat proteins
2. Structural organization of repeats
3. Repeat unit identification
4. Evolution through duplication
5. Structural symmetry
6. Functional roles of repeat proteins
7. Hydrophobicity and sequence patterns
8. Repeat classification
9. RepeatsDB
10. Disease-related repeat proteins
 1. What is a tandem repeat protein?
 2. How do tandem repeats arise during evolution?
 3. Why are tandem repeat proteins particularly common in eukaryotes?
 4. What biological functions are associated with repeat proteins?
 5. How can tandem repeats be identified from sequence or structure?
 6. What is structural symmetry in repeat proteins?
 7. Why do repeat proteins evolve rapidly?
 8. How are tandem repeats associated with disease?
 9. What is RepeatsDB and what information does it provide?
 10. How do tandem repeat proteins differ from globular proteins?

22 SB12 – Intrinsically Disordered Proteins

Table of Contents

1. Definition of intrinsically disordered proteins
2. Historical perspective
3. Structural disorder versus ordered proteins
4. Functional roles of disorder
5. Coupled folding and binding
6. Induced fit mechanisms
7. Signalling and regulation
8. Protein interaction networks
9. Experimental characterization of IDPs
10. Biological significance of disorder
 1. What is an intrinsically disordered protein (IDP)?
 2. Why can proteins lacking a stable structure still be functional?
 3. What is coupled folding and binding?
 4. What is induced fit?
 5. Why are IDPs frequently involved in signalling and regulation?
 6. Which sequence features are characteristic of IDPs?
 7. How do IDPs differ from globular proteins in terms of aminoacid composition?
 8. What advantages does structural disorder provide?
 9. Why are intrinsically disordered proteins difficult to study experimentally?
 10. Give examples of biological functions performed by IDPs.

23 SB13 – Disorder Prediction

Table of Contents

1. Low-complexity regions
2. SEG algorithm
3. NORS proteins
4. Amino-acid composition of disordered regions
5. Charge and hydrophobicity
6. Compositional bias
7. Evolution of intrinsic disorder
8. Sequence-based prediction methods
9. Benchmarking and CAID
10. Applications of disorder prediction
 1. What are low-complexity regions?
 2. How does the SEG algorithm identify low-complexity regions?
 3. What are NORS proteins?
 4. Which amino acids are enriched in intrinsically disordered regions?
 5. Why are hydrophobic residues generally depleted in disordered proteins?
 6. How does net charge contribute to disorder?
 7. What is compositional bias?
 8. Why is intrinsic disorder more prevalent in eukaryotes?
 9. How is disorder predicted directly from sequence?
 10. What are the strengths and limitations of disorder predictors?

24 SB14 – Molecular Dynamics

Table of Contents

1. Protein dynamics
2. Molecular dynamics simulations
3. Newtonian mechanics
4. Force fields
5. Potential energy functions
6. Simulation workflow
7. Solvation and boundary conditions
8. Conformational sampling
9. Analysis of trajectories
10. Strengths and limitations of MD
 1. Why is protein dynamics important for biological function?
 2. What is molecular dynamics (MD)?
 3. What is a force field?
 4. Which energetic terms are included in a typical force field?
 5. How are atomic trajectories generated in MD simulations?
 6. Why is explicit solvent often included in simulations?
 7. What is conformational sampling?
 8. What timescale limitations affect MD simulations?
 9. How can molecular dynamics help us understand protein function?
 10. What are the main limitations and sources of error in MD simulations?

25 SB15 – RING (Residue Interaction Networks)

Table of Contents

1. Graph representation of proteins
2. Residue interaction networks
3. Nodes and edges
4. Contact maps
5. Non-covalent interaction types
6. Network construction
7. Network topology
8. Centrality measures
9. Functional interpretation of networks
10. Applications of RING
 1. What is a residue interaction network (RIN)?
 2. Why can proteins be represented as graphs?
 3. What do nodes and edges represent in a residue interaction network?
 4. Which interaction types can define edges in a RIN?
 5. How is a contact map constructed?
 6. What information is lost when converting a protein structure into a graph?
 7. How can centrality measures identify important residues?
 8. How are residue interaction networks used to study mutations?
 9. What is the difference between atomic and residue-level interaction networks?
 10. What are the strengths and weaknesses of the RING approach?

26 SB16 – Embeddings and Protein Language Models

Table of Contents

1. Machine learning for proteins
2. Sequence representations
3. One-hot encoding
4. Multiple sequence alignments
5. Protein language models
6. Transformers
7. Self-supervised learning
8. Protein embeddings
9. Downstream prediction tasks
10. Applications and limitations of embeddings
 1. What is a protein language model (pLM)?
 2. Why is one-hot encoding insufficient for representing proteins?
 3. What is an embedding?
 4. Why can protein sequences be treated similarly to natural language?
 5. What is self-supervised learning?
 6. Why are transformer architectures effective for protein sequences?
 7. What biological information is captured by protein embeddings?
 8. How can embeddings be used to predict structure or function?
 9. What are the differences between embeddings and multiple sequence alignments?
 10. What are the main limitations of protein language models?